

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/01

Paper 1 Multiple Choice

21 September 2015

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

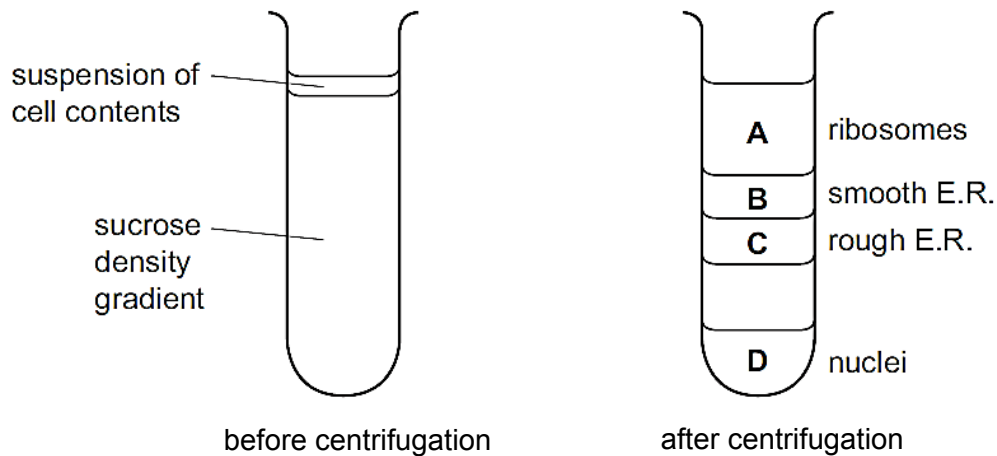
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **20** printed pages.



- 1 In zonal centrifugation, the suspension of cell contents is placed on top of a sucrose density gradient. The tube is then placed in a centrifuge and spun at high speed. The heavier particles will move towards the bottom of the tube faster than lighter particles, as shown below.

If a sample of intact prokaryotes had been added to a suspension of eukaryotic cell contents, where would you expect them to be found?



- 2 Which one of the following sequences of cell structures correctly describes the pathway that leads to the production, transport, and secretion of the protein thyroxine by thyroid gland cells?

- A Nucleus → ribosomes → endoplasmic reticulum → Golgi body → cell membrane
- B Nucleus → ribosomes → vesicles → endoplasmic reticulum → cell membrane
- C Ribosomes → mitochondria → Golgi body → vesicles → cell membrane
- D Ribosomes → endoplasmic reticulum → nucleus → Golgi body → cell membrane

- 3 Which of the statements about polysaccharides can be used to describe both amylopectin and cellulose?

- 1 adjacent glucose molecules are rotated by 180°
- 2 contains 1,4 glycosidic bonds
- 3 polymer of α-glucose

- A 2 only
- B 3 only
- C 1 and 2
- D 1 and 3

- 4 A student carried out four tests for biological molecules on a solution. The results are shown in the table.

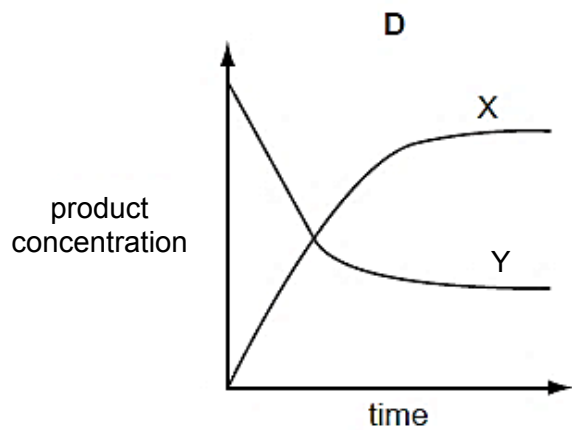
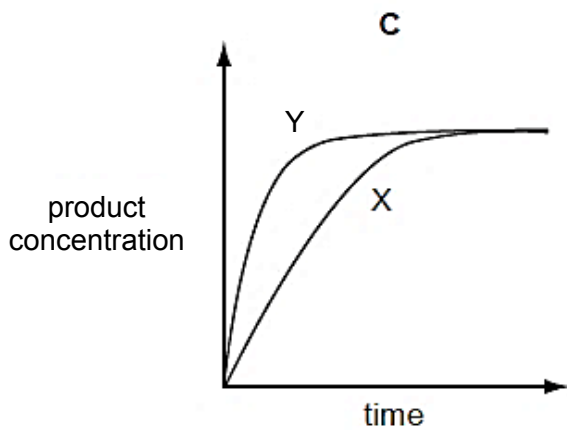
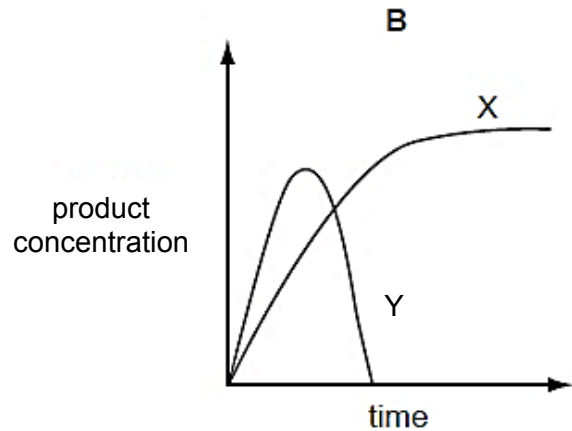
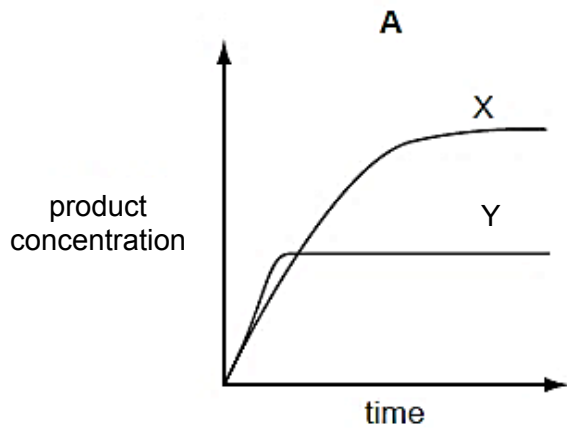
test	solution	observation
1	iodine solution	yellowish-brown
2	Biuret	purple
3	Benedict's	orange
4	emulsion	clear

The solution contain

- A collagen, sucrose.
 B collagen, glucose, sucrose.
 C collagen, fats, sucrose.
 D glucose, starch, fats.
- 5 Which features of collagen result in it having high tensile strength?
- 1 covalent bonds form between adjacent molecules
 - 2 each three-stranded molecule is held together by hydrogen bonds
 - 3 every third amino acid in the polypeptide is small
 - 4 the primary structure is held together by peptide bonds
- A 1 and 2
 B 1, 2 and 3
 C 1, 3 and 4
 D 2, 3 and 4

- 6 Two enzyme experiments were carried out. The first, experiment X, was carried out at a constant temperature of 37°C . During the second experiment, Y, the temperature was increased from 37°C to 80°C . All other factors were kept the same.

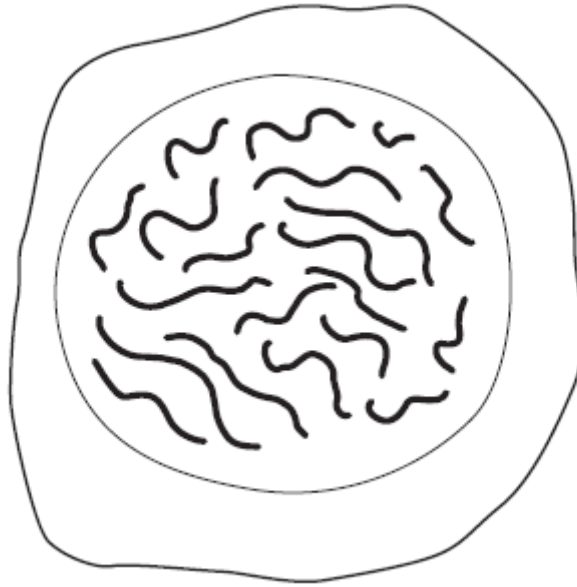
Which graph shows the results?



- 7 Which of the following statements describe non-competitive inhibitor of enzyme action?
- 1 Increasing the concentration of the enzyme's substrate will reduce its effect.
 - 2 It reduces the activation energy required for a reaction to take place.
 - 3 It reduces the maximum rate of reaction.

- A** 1 only
B 3 only
C 1 and 3 only
D 2 and 3 only

- 8 The diagram shows a cell of an organism formed by reductive division.



What is the diploid number for this organism?

- A 10
B 20
C 40
D 46
- 9 Which features of mitosis ensure that the genetic constitution of the cell is maintained?
- 1 the position of the chromosomes on the equator of the spindle
 - 2 the longitudinal division of the centromeres
 - 3 the DNA of the parent cells replicates before mitosis begins
 - 4 the pulling apart of the chromatids to opposite poles
- A 1, 2 and 3
B 1, 2, and 4
C 3 and 4
D 1, 2, 3 and 4
- 10 What is the maximum number of hydrogen bonds in a length of DNA containing 700 nucleotides?
- A 350
B 700
C 1050
D 2100

- 11** During semi-conservative replication of DNA in eukaryotic cells, the following processes occur.
- 1 Free nucleotides are hydrogen bonded to those on the exposed strand.
 - 2 Hydrogen bonds are broken between the complementary base pairs.
 - 3 Covalent bonds form between adjacent nucleotides on the same strand.
 - 4 The DNA double helix is unwound.

Which shows the correct order of some of the processes?

- A** 3 → 1 → 2 → 4
B 3 → 2 → 4 → 1
C 4 → 2 → 1 → 3
D 4 → 1 → 2 → 3
- 12** In eukaryotic cells, the termination of transcription occurs when a
- A** 'stop' codon is read in the DNA.
B polyadenylation signal is read in the DNA.
C 'stop' codon is synthesized in the mRNA.
D polyadenylation is synthesized in the mRNA.
- 13** The diagram shows the stages in the production of a polypeptide

DNA nucleotide sequence	
template strand	T A C G A C A A T C G C
mRNA sequence	A U G C U G U U A G C G
amino acid sequence	met leu leu ala

Which feature of the triplet code is illustrated by the information given?

- A** An amino acid can be coded for by more than one triplet.
B The triplet code is non-overlapping and is only read in one direction.
C The triplet code is universal for the DNA of all organisms.
D There are some triplets that code for 'start' and 'stop'.

- 14 The table shows some of the mRNA codons for some amino acids:

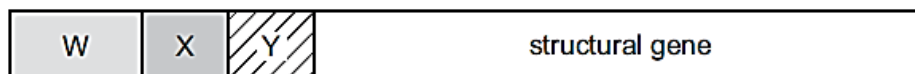
mRNA codon	amino acid
AUG	methionine
AAG	lysine
GGC, GGU	glycine
UUU, UUA	phenylalanine
AUA	isoleucine
UAA, UAG	--

A section of mRNA has the following codons:

<i>codon</i>		51	52	53	54	55	
<i>mRNA codon</i>		GGC	AAG	UUU	AUA	AAG	

Which one of the following changes to the mRNA section shown above would **not** result in the production of a different protein?

- A Deletion of the third U in codon 53
 B Insertion of U between codons 51 and 52.
 C Substitution of U for the second A in codon 52.
 D Substitution of U for C in codon 51.
- 15 The diagram represents a length of DNA from a prokaryote that includes a structural gene. Parts of the length of DNA are labelled W, X and Y. They have different functions in the control of transcription of the structural gene.



What identifies the functions of parts W, X and Y?

	W	X	Y
A	operator	regulator	regulator
B	promoter	regulator	operator
C	regulator	promoter	operator
D	promoter	operator	promoter

16 Which statements about bacterial genetic transfer are **not** correct?

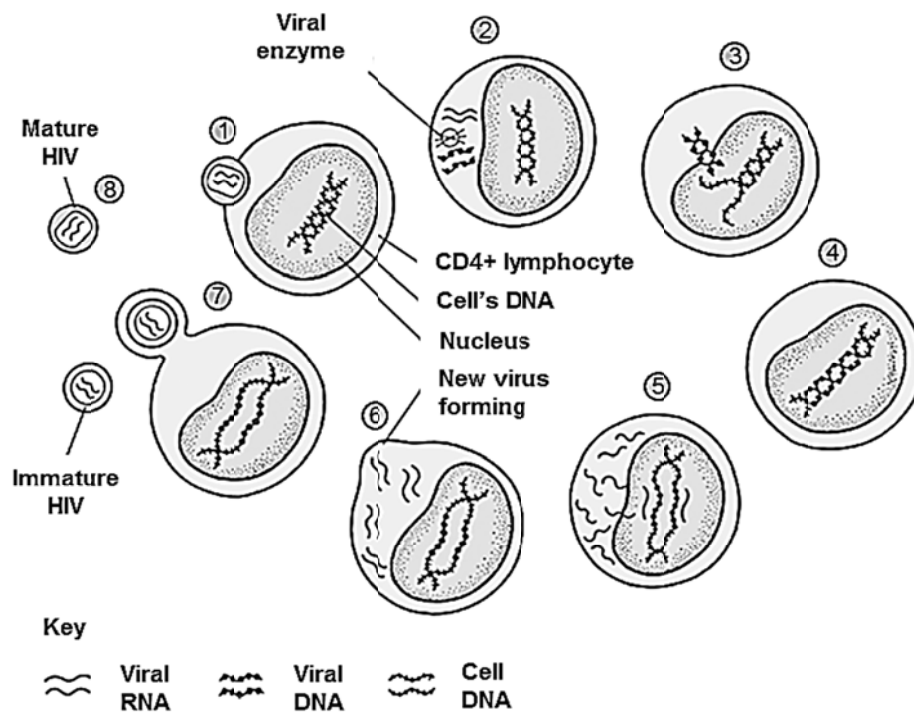
- 1 In transformation, only bacterial cells which possess competence factors in their chromosomes can take up naked DNA from the surroundings.
- 2 In generalised transduction, bacterial genes near the prophage insertion site on the donor chromosome can be transferred to the recipient cell via a T4 phage.
- 3 In conjugation, DNA replication of the F plasmid occurs in both donor and recipient cells.
- 4 Binary fission does not lead to variation in bacterial cells.

- A** 2 only
B 1 and 2
C 2 and 3
D 3 and 4

17 A prophage is a

- A** bacteria infected by a phage.
B phage infected by a bacteria.
C bacterial chromosome integrated into a phage chromosome.
D phage genome integrated into a bacterial chromosome.

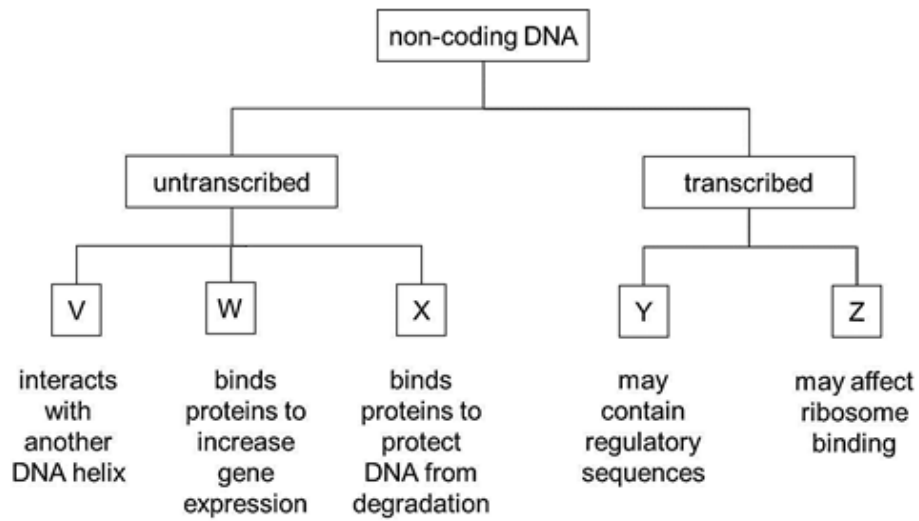
- 18 The diagram shows stages (1 to 8) of the reproductive cycle of a human immunodeficiency virus (HIV) in infecting a human CD4⁺ lymphocyte.



Which row correctly explains how the reproductive cycle of HIV can be arrested at the various stages?

	stage 1	stage 2	stage 4	stage 7
A	drugs that bind to gp120	inhibition of viral replicase	mutation in the HIV genome	inhibition of exocytosis
B	restriction endonucleases to cut the viral genome	inhibition of viral replicase	inhibition of integrase	inhibition of budding
C	drugs that bind to gp120	inhibition of reverse transcriptase	inhibition of integrase	inhibition of budding
D	restriction endonucleases to cut the viral genome	inhibition of reverse transcriptase	mutation in the HIV genome	inhibition of exocytosis

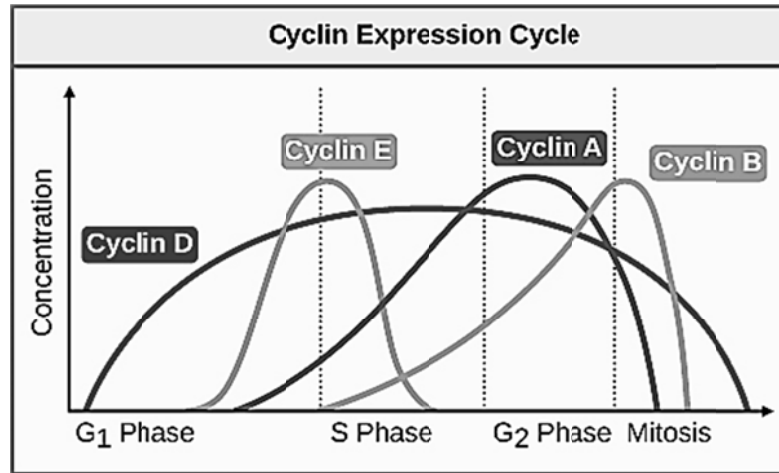
- 19 The diagram shows the classification of several portions of eukaryotic DNA that do not encode RNA or protein.



Which combination correctly identifies the type of eukaryotic DNA?

	V	W	X	Y	Z
A	centromere	intron	telomere	enhancer	5' UTR
B	centromere	enhancer	telomere	intron	5' UTR
C	intron	enhancer	telomere	5' UTR	centromere
D	telomere	enhancer	intron	5' UTR	centromere

- 20** Cyclins are regulatory proteins that associate with cyclin-dependent kinases (CDKs) to control the different stages of the cell cycle. The right type and the correct amount of cyclins and CDKs must be present at the different stages to ensure regulation of the cell cycle. The concentrations of the different cyclins are shown in the diagram.



How could the levels of the different cyclins be regulated?

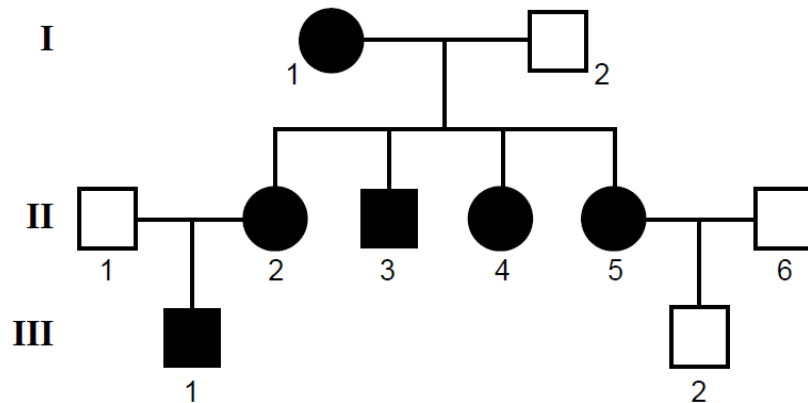
- 1 formation of heterochromatin
- 2 binding of repressor to operator
- 3 length of mRNA poly(A) tail
- 4 ubiquitination of cyclins

- A** 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

- 21** Which of the following scenarios have the highest risk of cancer?

	proto-oncogene	tumour-suppressor gene
A	gain of function in one allele	loss of function in both alleles
B	gain of function in both alleles	loss of function in one allele
C	loss of function in both alleles	gain of function in one allele
D	loss of function in one allele	gain of function in one allele

- 22 Phenylketonuria (PKU) is inherited as an autosomal recessive condition. In the following pedigree, shaded individuals have this trait.



Which of the following can be best concluded from the pedigree?

- A I2 is heterozygous
 - B II1 is homozygous
 - C II5 is heterozygous
 - D III2 is homozygous
- 23 In domestic poultry, the allele for white feathers (W) is dominant to the allele for dark feathers (w), and the allele for frizzled feathers (F) is dominant to the allele for normal feathers (f).

Poultry which were homozygous for dark, frizzled feathers were bred with poultry bearing white, normal feathers. The offspring of this cross were crossed with homozygous recessive individuals and the following results obtained:

Phenotype	Number of offspring
white, frizzled feather	18
dark, frizzled feather	63
white, normal feather	63
dark, normal feather	13

Which one of the following statements are consistent with these results?

- A The genes are found on 2 separate pairs of homologous chromosomes.
- B The genes are linked on the same chromosome and allele W is linked to allele w.
- C The genes are linked on the same chromosome and allele W is linked to allele F.
- D The genes are linked on the same chromosome and allele w is linked to allele F.

- 24 An insect collection contains 102 specimens of a species of butterfly. This specimen is sexually dimorphic, meaning that the males and females look different from each other. A student examined the specimens and collected the following data.

Observation	Frequency
Blue wing colour (male)	64
Brown wing colour (female)	38
Wing span 35-37 mm	15
Wing span 37-39 mm	68
Wing span 39-41 mm	19

How should this variation be classified?

	continuous variation	discontinuous variation
A	sexual dimorphism	colour
B	colour	sexual dimorphism
C	sexual dimorphism	wingspan
D	wingspan	sexual dimorphism

- 25 The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A controlled experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

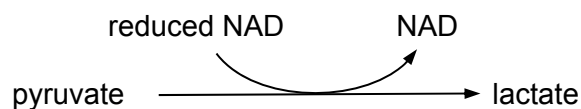
The following results were obtained:

light intensity (% of full sunlight)	concentration of CO ₂ in air after flowing over leaves (ppm)
75	201
50	200
25	254
10	280

Which one of the following statements is **not** consistent with these results?

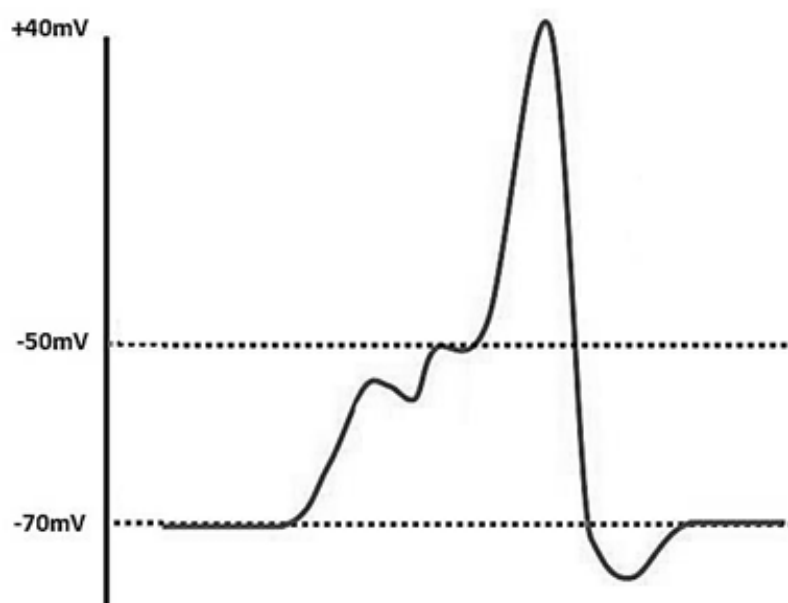
- A At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.
- B In the dark, the concentration of carbon dioxide in the air after it had flowed over leaves would be at least 280 ppm.
- C At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.
- D At a light intensity of 10% of full sunlight, carbon dioxide concentration is limiting.

- 26 Lactate dehydrogenase catalyses the following reaction:



What would be the effect of inhibition of lactate dehydrogenase in a mammalian cell under anaerobic conditions?

- A A decrease in glycolysis, due to the lack of NAD.
 - B A decrease in cell pH, due to the accumulation of lactic acid.
 - C An increase in ATP production, due to increased amounts of reduced NAD.
 - D An increase in the activity of Krebs cycle, due to increased amounts of pyruvate.
- 27 The diagram below shows the membrane potential of a post-synaptic neurone.



Which of the following statements are likely to correspond to the diagram?

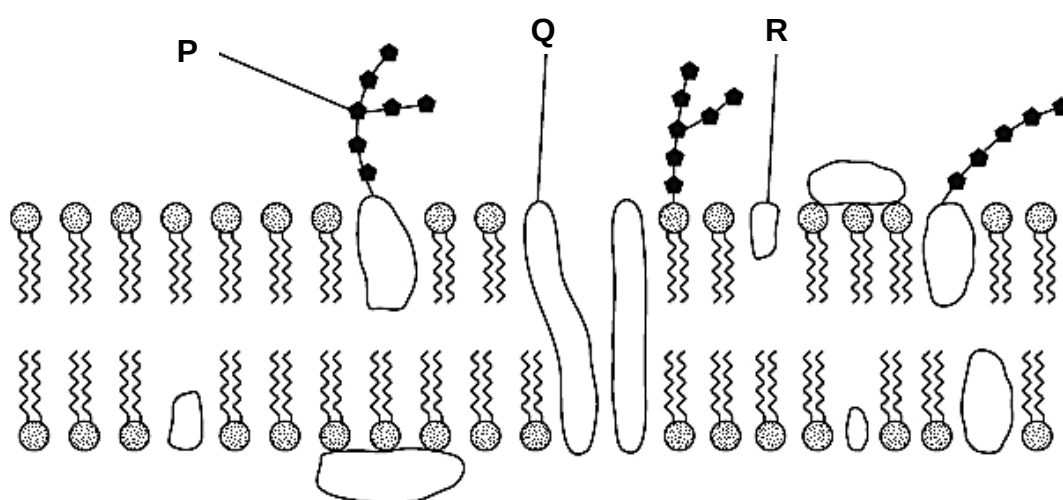
- 1 repeated release of neurotransmitters from a single pre-synaptic terminal
 - 2 simultaneous release of neurotransmitters from several pre-synaptic terminals
 - 3 potassium ion channels on post-synaptic membrane are opened
 - 4 sodium ion channels on post-synaptic membrane are opened
- A 1 and 3
 - B 1 and 4
 - C 2 and 3
 - D 2 and 4

28 Which of the following statements are true about the signal transduction pathways?

- 1 They enable different cells to respond appropriately to the same signal.
- 2 They help cells use up phosphate generated by ATP breakdown.
- 3 They can amplify a signal.
- 4 Variations in the signal transduction pathways can enhance response specificity.

- A** 1 and 3
B 2 and 4
C 1, 2 and 3
D 1, 3 and 4

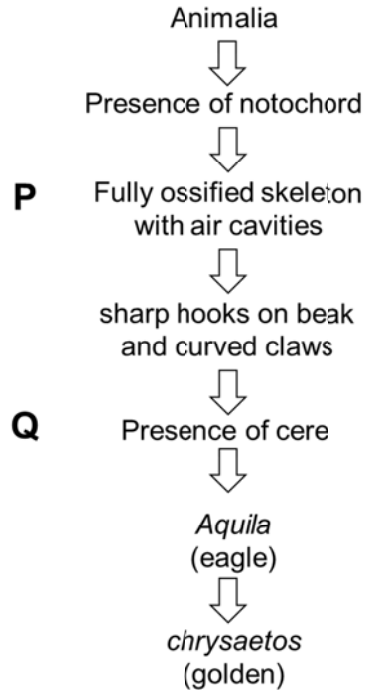
29 The diagram shows part of a cell surface membrane.



What is the correct function for each of the structures labelled?

	regulates membrane fluidity	forms hydrogen bonds with water to stabilise membrane	transports ions and large polar molecules
A	P	R	Q
B	P	Q	R
C	Q	R	P
D	R	P	Q

- 30 The diagram shows features used in classifying the golden eagle.

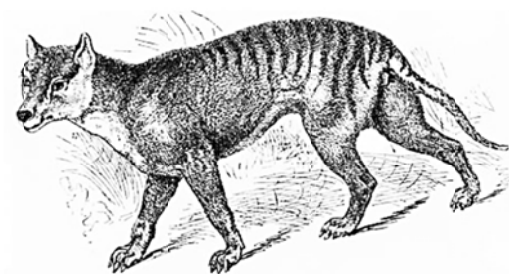


Which taxa do P and Q refer to?

	P	Q
A	class	genus
B	class	family
C	genus	class
D	species	genus

- 31 The Tasmanian tiger (now extinct) and the American grey wolf evolved independently of each other, but show similar physical structures and hunting behaviours.

The similarities between the two organisms are most likely a result of



Tasmanian tiger



American grey wolf

- A allopatric speciation.
- B convergent evolution.
- C genetic drift.
- D homology.

32 What are the advantages and limitations of the polymerase chain reaction?

- 1 The length of DNA to be amplified is approximately 5kb.
- 2 The amount of DNA required is small.
- 3 There is an exponential increase in the DNA content during each cycle.
- 4 Specific primers are required for specific target sequences.

	Advantages	Limitations
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

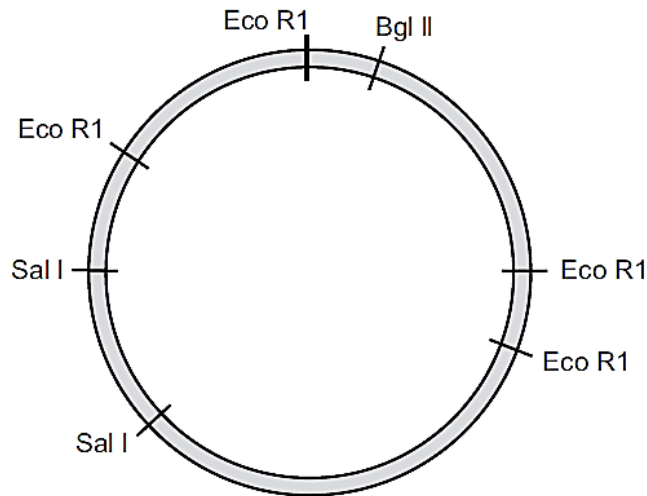
33 DNA fingerprints were obtained from four sources.

Crime scene sample	Victim	Suspect A	Suspect B
==	—		—
==		==	
—	—		
==		—	—
==		—	—
==	==	—	—

The sample taken from the crime scene contains DNA from the victim and

- A** an unknown individual.
- B** Suspect A.
- C** Suspect B.
- D** Suspect A and Suspect B.

- 34 The following diagram indicates the cutting sites of three different restriction enzymes on a particular bacterial plasmid.



How many fragments of DNA would be obtained if the plasmid was completely digested with the restriction enzyme *Eco R1*?

- A 2
B 3
C 4
D 7
- 35 Genes P, Q, R and S occur on the same chromosome. The table shows the recombination frequencies.

	recombination frequency (%)
between P and Q	46
between P and R	8
between R and Q	54
between Q and S	13
between R and S	41

Which of the following represents the correct order of genes on the chromosome?

- A P – Q – R – S
B P – R – S – Q
C R – P – S – Q
D R – S – Q – P

- 36** Which of the following statements regarding embryonic stem cells and hematopoietic stem cells is true?
- A** As embryonic stem cells develop, they turned into hematopoietic stem cells as they lose their ability to differentiate into all types of cells.
 - B** Both stem cells are derived from the zygotic stem cells with the hematopoietic stem cells having a lowered telomerase activity compared to the embryonic stem cells.
 - C** Embryonic stem cells have more genes than hematopoietic stem cells and thus are able to form more types of cells.
 - D** Under normal conditions, embryonic stem cells express more of their genes compared to the hematopoietic stem cells.

- 37** Which of the following correctly describes germline gene therapy?
- A** It could potentially correct a genetic defect in an affected individual only.
 - B** It could potentially correct a genetic defect in an affected individual and all of his or her descendants.
 - C** It could potentially correct a genetic defect in the offspring of an affected individual only.
 - D** It could potentially correct a genetic defect in the parent of an affected child.

- 38** Pollen or anther culture involves the growing of haploid plantlets from the male gametes of the plant.

Embryo culture involves the removal of the embryo after successful fertilization of an ovule with pollen and the subsequent culture of the embryo to form a plantlet.

Which of the following is a similarity between the two plant tissue culture methods mentioned above?

- A** Homozygous plants can be produced in large numbers from calli.
 - B** Large number of genetically identical plants are obtained.
 - C** Resultant plants will be sterile.
 - D** Variation in the level of plant growth regulators leads to the formation of specific plant parts.
- 39** Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

- A** salmon providing eggs for treatment
- B** trout providing reproductive organ tissue
- C** young triploid salmon
- D** young trout

- 40** Which of the following statements below best represent a social implication of genetically modified (GM) crop plants?
- A** There is evidence that genes for herbicide resistance have spread from GM oilseed rape to nearby weed species although there is no evidence that these hybrids have persisted in the environment.
 - B** In some countries, government regulations have been put in place to prevent farmers planting GM crops close to non-GM and organic crops to prevent GM pollen from reaching the crops.
 - C** In some studies, the number of beneficial insects has been reduced in areas where insect resistant GM Bt corn has been grown, however in other studies, there has been no effect or the number of beneficial insects has increased.
 - D** Pressure placed by people on a multinational GM seed company has caused them to pledge that they will never develop a 'suicide' gene that prevents farmers from growing seed collected from last year's crop.

2015 JC2 Prelim 2
9648 H2 Biology
Paper 1
Suggested Answers

Qn	Ans
1	D
2	A
3	A
4	B
5	B
6	A
7	B
8	C
9	D
10	C

Qn	Ans
11	C
12	D
13	A
14	D
15	C
16	A
17	D
18	C
19	B
20	D

Qn	Ans
21	A
22	A
23	D
24	D
25	D
26	A
27	B
28	D
29	D
30	B

Qn	Ans
31	B
32	C
33	D
34	C
35	C
36	A
37	C
38	D
39	C
40	B